DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	MANAGEMENT OF NEONATAL HYPOCALCEMIA
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

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CONTENTS

1. PURPOSE	4. POLICY	7. REFERENCE/CROSS-REFERENCES
2. DEFINITION & ABBREVIATION	5. PROCEDURE	8. ATTACHMENT
3. RESPONSIBILITY	6. MEASUREMENT	9. REVISION CLAUSE

1. PURPOSE

1.1. To provide guidelines on how to assess and manage newborn with neonatal hypocalcemia.

2. DEFINITION & ABBREVIATION

2.1. **Hypocalcemia** – defines as a total serum calcium concentration of less than 8 mg/dL in term infants or less than 7 mg/dL in preterm infants, and it can lead to significant neurological symptoms if not addressed.

2.1.1. **Hypocalcemia in neonates can be classified into two types**:

2.1.1.1. **Early-Onset Hypocalcemia**: Occurs within the first 2 to 3 days of life often resolving spontaneously. It is commonly associated with conditions such as prematurity, being small for gestational age, maternal diabetes, and perinatal asphyxia.

2.1.1.2. **Late-Onset Hypocalcemia**: Develops after the first week of life and is less common. It can be caused by dietary factors, such as excessive phosphate intake from cow's milk or formula, or issues with parathyroid.

2.2. **IV** – intravenous

2.3. **Ca** - Calcium

2.4. **HIS (Hospital Information System)** - an element of health informatics that focuses mainly on the administrational needs of hospitals.

2.5. **MCC (Medica Cloud Care)** – the HIS that is utilized in PMSH for patient's data.

2.6. **EMR** –Electronic Medical Record

2.7. **NICU** – Neonatal Intensive Care Unit

3. RESPONSIBILITY

3.1. **Head of NICU**

3.1.1. Ensures the strict adherence of all staff to this policy and procedures.

3.2. **Physicians (Consultant, Specialists and Residents)**

3.2.1. Provides the orders in the physician order's sheet in the EMR.

3.3. **Nurses (Charge Nurse and Staff Nurse)**

3.3.1. Verify and implement physician's order.

3.3.2. Assess, reassess and monitor patients with hypocalcemia.

4. POLICY

4.1. Physicians will follow the recommended treatment and considerations in the management of

hypocalcemia as stated below:

4.1.1. Starting management of hypocalcemia when newborn's serum calcium level is Ca < 1.9 mmol/L or < 7 mg/dl. Normal range of Ca = 2.2 – 2.7 mmol/L (ionized Ca ++ normal range = 1.14 – 1.29). Corrected Ca = 0.02 x (40 – measured albumin) + measured Ca.

4.2. Never give IV push (calcium) or thru subcutaneous.

4.3. All patients on IV calcium should be on cardiac monitor.

4.4. Monitor IV site q1-2H for: extravasations (may cause burns, calcifications), various irritation, phlebitis, thrombosis

4.5. Check for serum magnesium and Vitamin D levels as they were associated factors for neonatal hypocalcemia

5. PROCEDURE

5.1. The physician and nurse on duty will assess patients with potential risk and predisposing factors to develop hypocalcemia which are as follows:

5.2. Patient with confirmed hypocalcemia will receive appropriate treatment as follows:

5.2.1. 1 mmol/L Ca (of any Ca solution) = 2 mEq = 40 mg elemental Ca (depend on the type of Ca solution)

5.2.2. 1 ml Ca gluconate 110% = 9.3 mg elemental Ca = 100 mg Ca gluconate salt = 0.23 mmol/L = 0.48 mEq

5.2.3. 1 ml Ca chloride 10% = 27 elemental Ca = 0.68 mmol = 1.3 Ca gluconate is the preferred Ca solution; Ca chloride should be avoided if possible.

5.3. **For patients with Symptomatic Hypocalcemia (seizure, tetany, stridor):**

5.3.1. 1 ml Ca chloride 10% = 27 mg elemental Ca = 0.68 mmol = 1.3 mEq Ca gluconate is the preferred Ca solution, Ca chloride should be avoided if possible.

5.3.2. Loading dose (Slow IV bolus): dose 5-10 mg elemental Ca/kg = 0.1 mmol/kg = 0.5 ml/kg of Ca gluconate 10% diluted 1:4, give slowly over 30 minutes, repeat if needed followed by:

5.3.3. Continuous infusion: dose – 2 mg elemental Ca/kg/hr = 0.5 mmol/kg/hr use Ca gluconate 2% (add 10 ml of 10% Ca gluconate + 40 ml NS or D5% NS to make Ca gluconate 2%) 1 ml Ca gluconate 2% = 2 mg elemental Ca = 0.04 mmol run at rate 1ml/kg/hr (this will give 2 mg elemental Ca/kg/hr) 0.05 mmol/kg/hr = 2 mg elemental Ca/kg/hr = e.g. if Wt = 10 kg: give 10ml/hr of 2% Ca gluconate.

5.3.4. Monitor serum Ca q6hrs & adjust Ca infusion rate accordingly.

5.3.5. Decrease Ca infusion rate by ½ (don't stop rapidly) if serum Ca 2.0 mmol

5.3.6. If serum Ca reached 2.2 mmol stop Ca infusion recheck serum Ca q6h

5.4. **For patients with Asymptomatic Hypocalcemia**

5.4.1. Give oral calcium 50 – 150 mg elemental Ca/kg/day ÷ q6h, use Ca Carbonate or Ca Sandoz Effervescent or syrup

5.4.2. Start 1-aplha drops 0.1 -0.2 mcg/kg/day q12h (maximum 1-2 mcg/day = 10-20 drops/day)

5.4.3. Monitor serum Ca & P04 once daily, & wean the above high doses accordingly oral Ca preparations:

5.4.3.1. Ca Carbonate tablets (Caltrate): 1 tablet contain 600 mg elemental Ca (better bioavailability)

5.4.3.2. Ca Sandoz Syrup: 1 ml = 21.6 elemental Ca = 8.1 mmol = 16.2 mEq (as Ca glubionate + Ca Lactate)

5.4.3.3. 1 drop of 1 –alpha contain 0.1 mcg

5.5. Physician will enter orders in the doctors order sheet in MCC.

5.6. Nurse on duty will verify physician's order and implement treatment as prescribed.

5.7. If hypocalcemia persists after 48 hours, look for underlying causes, (e.g. hypoparathyroidism, magnesium deficiency)

5.8. Re-assessment of patients receiving treatment of hypocalcemia shall be performed.

5.9. Document all the procedures done in the patient's electronic medical record (EMR).

6. MEASUREMENT

6.1. 100% of the staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0027(v2): Management of Neonatal Hypocalcemia, 03/07/2023, p. 1-4

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10, NICU.12 p.126-127

7.3. Harriet Lane 17th edition, 2009

7.4. NGHA-Parental Therapy Manual, 2006

7.5. HSC Handbook of Pediatrics, 2004 10th edition

7.6. Pediatric Endocrinology by Sperling 2nd edition, 2002

7.7. Pocket pediatrics, Ocallaghan and Stephenson, 2001

8. ATTACHMENT

8.1. None

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.

9.2. Change of code from DPP-NICU-0027 to DPP-NICU-0012

9.3. Modification of definition and abbreviation